

INTRODUCTION TO PVSYST SOFTWARE
LABORATORY SESSION 02
ELECTRICAL INSTALLATION DESIGN
EC9180

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2022 / E / 052

GROUP EG-06

SEMESTER 7

28TH AUG 2026

INTRODUCTION

Grid connected solar PV design in tropical climates demands high fidelity simulation to bridge the gap between nominal STC ratings and actual field performance. Simplified estimations based on peak sun hours ignore hourly meteorological variability, 3D near shading inter row effects, temperature derating, inverter clipping, and electrical mismatch caused by partial shading. PVsyst addresses these limitations through hourly time step simulation using Meteonorm meteorological data, Perez transposition models, and Monte Carlo style raytracing for 3D obstructions.

This laboratory designs a **100 kWp rooftop grid-connected system** for **Ariviyal Nagar, Kilinochchi, Sri Lanka** (Lat 9.3840°N, Long 80.4087°E, Alt 29 m). Kilinochchi receives ~1910 kWh/m² annual global horizontal irradiation with low seasonal variation. At this near equatorial latitude (~9.4°N), the solar path is nearly overhead during equinoxes, favouring low tilt arrays but simultaneously increasing sensitivity to inter row shading if module height or tilt is excessive.

The system quality is quantified by the **Performance Ratio (PR)**:

$$PR = \frac{Y_f}{Y_r} = \frac{E_{Grid}/P_o}{H_i/G_o}$$

where E_{Grid} is the net annual energy injected into the grid (kWh), P_o is the installed nominal DC peak power at STC (kWp), H_i is the total global incident irradiation on the tilted collector plane (kWh/m²), and $G_o = 1 \text{ kW/m}^2$ is the STC reference irradiance.

Near-shading losses are bifurcated into: - **Linear Shading Loss:** A purely geometric attenuation of beam and diffuse irradiance striking the array surface. - **Electrical Partition (Mismatch) Loss:** An additional power penalty arising from irradiance imbalance across module strings, causing bypass diode conduction and shifting the array away from its true maximum power point (MPP).

OBJECTIVES

1. **Project & Meteo Setup:** Configure a grid connected project in PVsyst and import Meteonorm 9.0 synthetic weather data for Kilinochchi.
2. **Component Selection & String Sizing:** Size a 100 kWp (+5%) DC array using Generic 550 Wp monocrystalline modules and a 100 kWac inverter, achieving a DC/AC ratio between 0.95–1.15.
3. **3D Near Shading Scene:** Construct the physical environment including the main building, adjacent obstruction, and rooftop cylindrical vent.
4. **Linear Shading Simulation:** Run the baseline simulation (landscape, 9.5° tilt, 2.5 m pitch) using the linear shading model; record E_{Grid} , PR, and the Loss Tree.
5. **Electrical Partition Simulation:** Re-configure near-shading to “According to Strings” with portrait arrangement (partition Y=2, X=1) and quantify the incremental electrical mismatch loss relative to the linear baseline.
6. **Pitch Variation Analysis:** Increase inter-row pitch from 2.5 m to 3.2 m (landscape) and evaluate the impact on Ground Coverage Ratio (GCR), shading loss factor, and total energy yield.

7. **Tilt Trade Off Analysis:** Increase array tilt from 9.5° to 18.0° (at 3.2 m pitch) and analyse the trade-off between transposition gain and increased inter-row shading.
8. **Loss Tree & Maintenance Strategy:** Deconstruct the PVsyst Loss Tree parameters (thermal coefficient U_c , ohmic wiring, mismatch, soiling) and propose field maintenance strategies to mitigate long-term PR degradation.

METHODOLOGY

1. Project & Meteorological Site were configured

- **Software:** PVsyst V8.1.5 (Evaluation Mode)
- **System Type:** Grid Connected
- **Location:** Kilinochchi, Sri Lanka
- **Coordinates:** 9.3840°N, 80.4087°E, 29 m altitude
- **Time Zone:** UTC+5.5
- **Weather Dataset:** Meteonorm 9.0 dll, Sat=100% – Synthetic
- **Albedo:** 0.20

2. Component Selection were selected & String were sized

Table 01: Component Specifications

Parameter	Specification
Target Installed Capacity	100 kWp (+5%)
PV Module	Generic 550 Wp Monocrystalline (Si-mono), Twin half-cells bifacial
Module Dimensions (typical)	~2279 mm × 1134 mm × 35 mm (landscape)
Inverter	Generic 100 kWac Grid-Connected Inverter
Inverter MPPT Range	180 – 1000 V
Number of MPPT Inputs	2 (50% allocation, no power sharing)
String Configuration	10 strings × 19 modules in series
Total Modules	190 units
Nominal DC Power (P_{nom})	105 kWp
DC/AC Ratio (P_{nom} ratio)	1.05
String MPP Voltage (50°C)	~718 V
String MPP Current (50°C)	~13.4 A
Total Array Current	~134 A

3. 3D Shading Environment was constructed

Table 2: 3D Scene Elements

Element	Dimensions	Coordinates / Position
Main Building Roof	Length 50.0 m, Width 30.0 m, Height 8.0 m	Origin (0, 0, 0)
Adjacent Building	Length 15.0 m, Width 15.0 m, Height 12.0 m	5.0 m East of Main Building
Rooftop Obstruction	Cylinder, Radius 1.5 m, Height 2.0 m, 18 segments	Centre of Main Roof
PV Shed Array	5 sheds, initial pitch 2.50 m	Mounted on Main Roof

4. Baseline Linear Shading Simulation was executed

Table 3: Array Configuration Parameters

Parameter	Task 2 (Baseline)	Task 4 (Electrical)	Task 6 (Pitch ↑)	Task 7 (Tilt ↑)
Module Orientation	Landscape	Portrait	Landscape	Landscape
Modules in Length (X)	19	19	19	19
Modules in Height (Y)	2	2	2	2
Number of Sheds	5	5	5	5
Pitch N–S	2.50 m	2.50 m	3.20 m	3.20 m
Tilt	9.5°	9.5°	9.5°	18.0°
Azimuth	0°	0°	0°	0°
Shading Model	Linear (Fast table)	According to Strings, Electrical 100%	Linear (Fast table)	Linear (Fast table)

PRACTICAL TASKS & EVALUATION

1. String Layout Matching

The array comprises **190 modules** arranged as **10 parallel strings of 19 series-connected modules**. At STC, each string produces $550\text{ Wp} \times 19 = 10,450\text{ Wp}$. Ten strings yield a total installed DC capacity of **104.5 kWp**, reported by PVsyst as **105 kWp**. The inverter nominal AC power is **100 kWac**, giving a DC/AC ratio of **1.05**, which satisfies the IEC 62548 design window of 0.95–1.15. The string MPP voltage at 50°C (~718 V) lies centrally within the inverter MPPT range (180–1000 V), ensuring full power harvesting across all ambient conditions. The two MPPT inputs are allocated 50% of the strings each (5 strings per MPPT), with **no power sharing** between inputs a critical constraint later exposed in the electrical partition analysis.

PV Array Characteristics			
PV module		Inverter	
Manufacturer	Generic	Manufacturer	Generic
Model	Mono 550 Wp Twin half-cells bifacial (Original PVsyst database)	Model	100 kWac inverter (Original PVsyst database)
Unit Nom. Power	550 Wp	Unit Nom. Power	100 kWac
Number of PV modules	190 units	Number of inverters	2 * MPPT 50% 1 unit
Nominal (STC)	105 kWp	Total power	100 kWac
Modules	10 units x 19 In series	Operating voltage	180-1000 V
At operating cond. (50°C)		Pnom ratio (DC:AC)	1.05
Pmpp	95.9 kWp	No power sharing between MPPTs	
U mpp	718 V		
I mpp	134 A		
Total PV power		Total inverter power	
Nominal (STC)	105 kWp	Total power	100 kWac
Total	190 modules	Number of inverters	1 unit
Module area	491 m²	Pnom ratio	1.05
Cell area	453 m²		

Figure 1: PV Array Characteristics

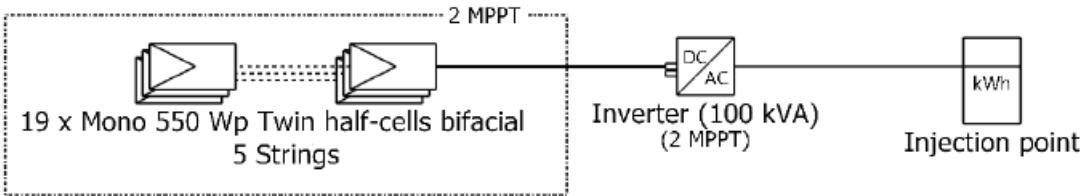


Figure 2: Single line diagram showing 19 module strings, 5 strings per MPPT, 100 kVA inverter, and grid injection point

2. Linear Shading Simulation

The baseline simulation employs the **Linear Shading (Fast Table)** method, which calculates the geometric reduction in beam and diffuse irradiance without modelling electrical mismatch.

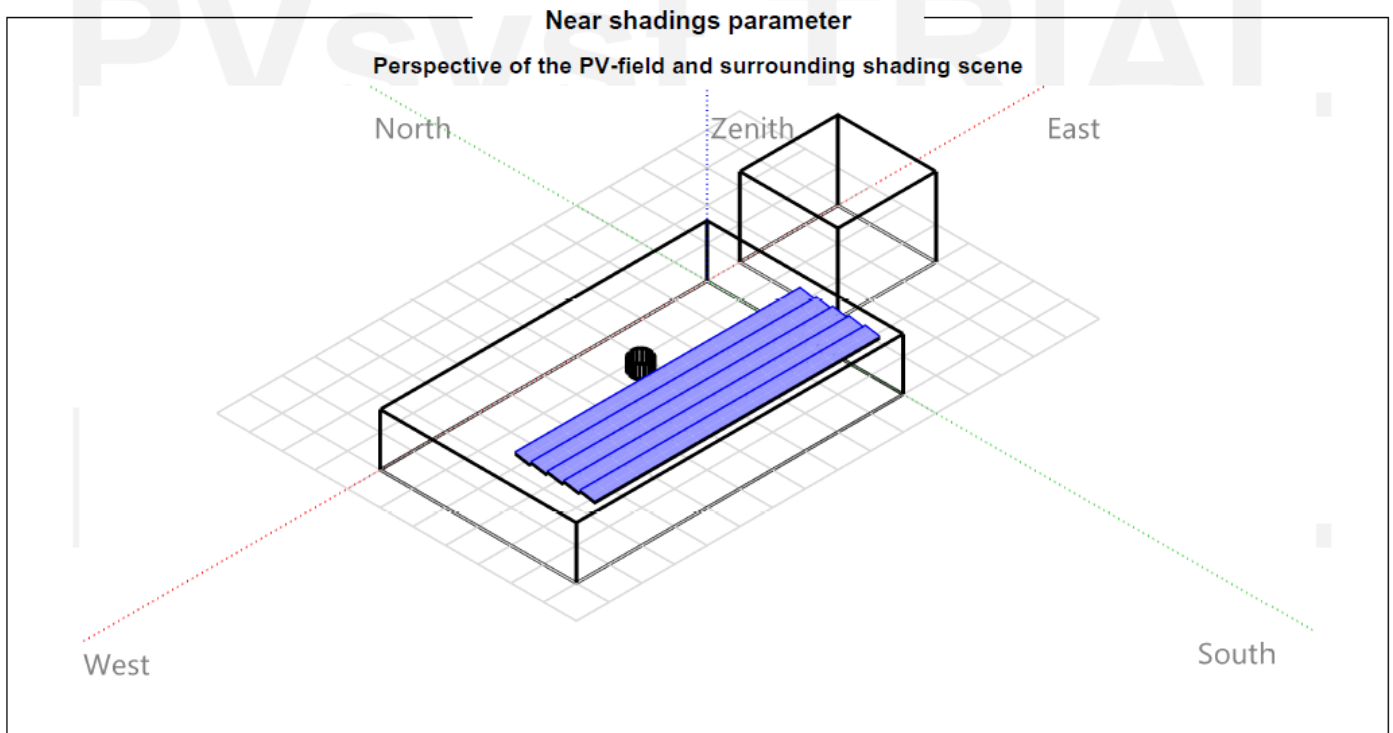


Figure 3: 3D perspective view of the PV field

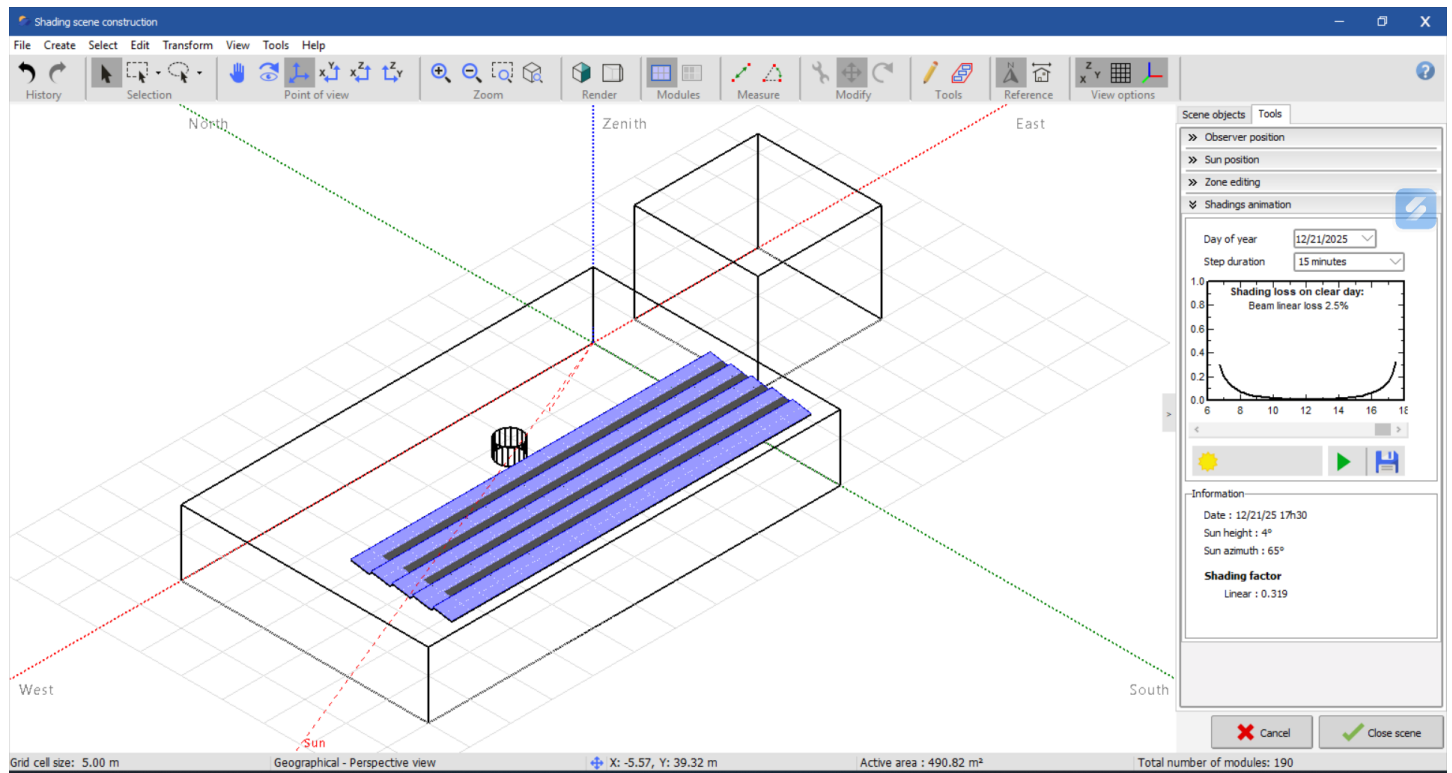


Figure 4: Simulation results

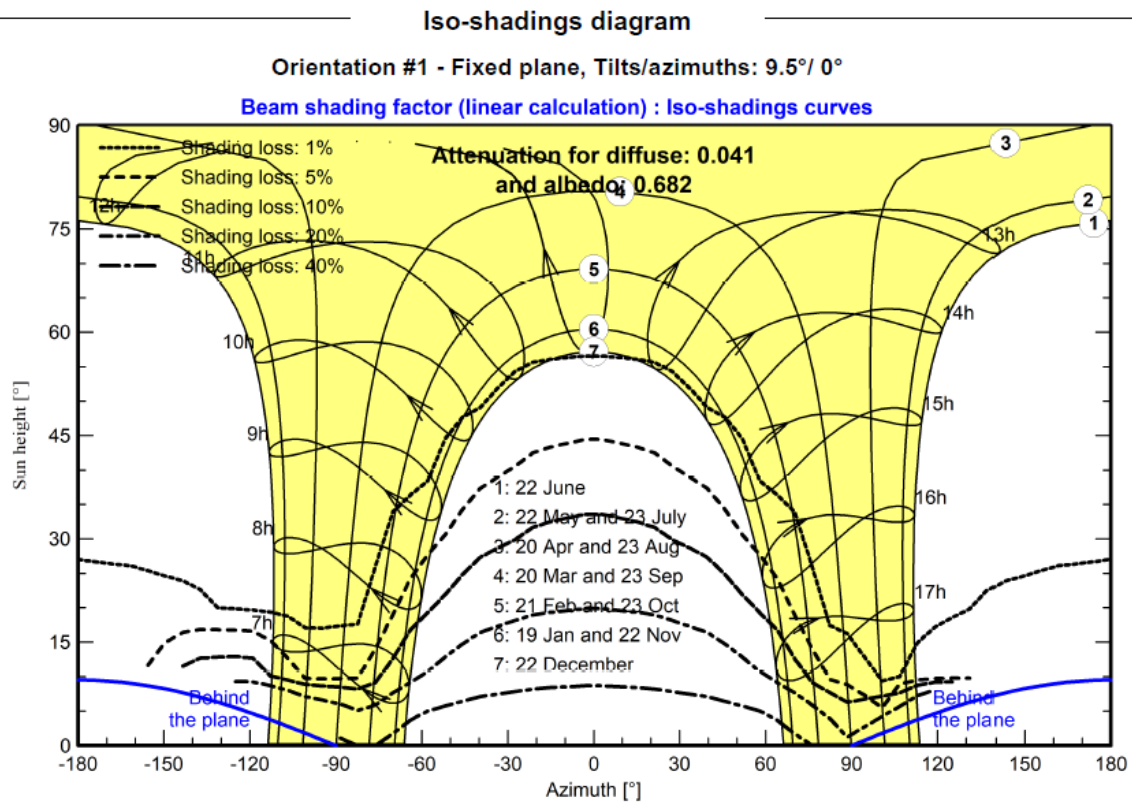


Figure 5: Iso-shading diagram for (9.5°/0°)

Table 04: Baseline Linear Shading Results

Metric	Value	Interpretation
Global Horizontal Irradiation (G_h)	1910.5 kWh/m ² /year	Kilinochchi solar resource
Global Incident ($G_{i,inc}$)	1922.5 kWh/m ² /year	Transposition gain (+0.6%)
Near-Shading Irradiance Loss	-1.64%	Minor inter-row loss at 2.5 m pitch
IAM Factor on Global	-2.18%	Incidence angle reflection loss
Effective Global ($G_{i,eff}$)	1849.8 kWh/m ² /year	Usable irradiance on collectors
Array Virtual Energy at MPP	169,199 kWh/year	DC energy before inverter
Energy Injected to Grid (E_{Grid})	166,475 kWh/year	Net annual AC yield
Specific Production	1593 kWh/kWp/year	Yield per installed kWp
Performance Ratio (PR)	82.86%	Excellent for tropical rooftop

Power Density Analysis: - Total installed power: 190 modules $\times$ 550 Wp = 104.5 kWp ; Roof area: 50 m $\times$ 30 m = 1500 m² ; Power density: 104,500 W / 1500 m² = **69.7 Wp/m²**

Uniformity & Shading Limit: The iso shading diagram reveals a **shading limit angle of 59.6°**. Contours intrude only marginally into the solar path, confirming that the 2.5 m pitch is adequate for the low 9.5° tilt in landscape orientation. The GCR (shading metric) is **92.3%**, indicating a dense packing that still avoids significant shading.

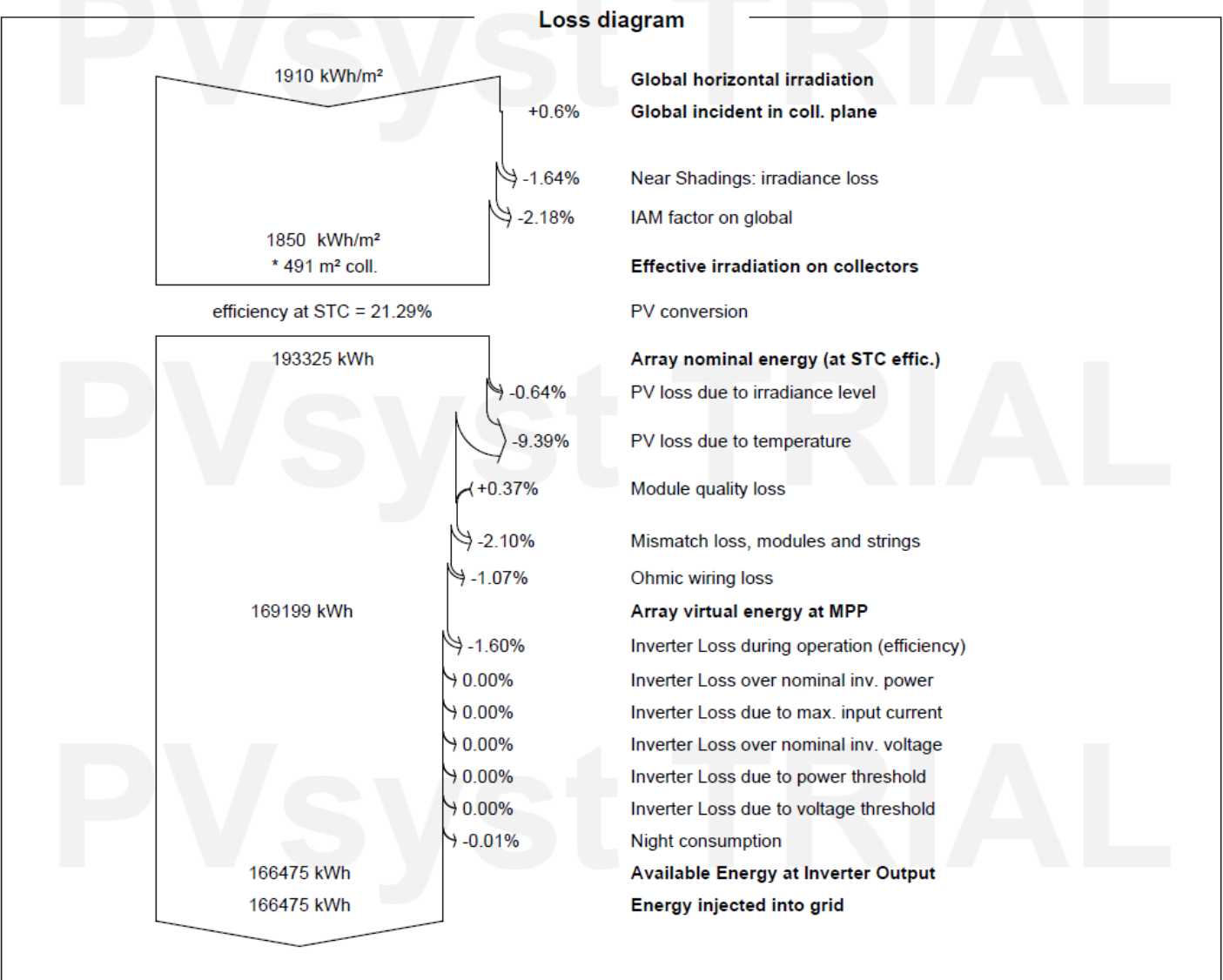


Figure 6: Loss Diagram

3 & 4. Electrical Partition Simulation (Portrait Arrangement)

Re configuring the near shading model to “According to Strings” with **Electrical Effect 100%** and portrait partition ($Y=2, X=1$) fundamentally alters the array geometry. In portrait orientation, the module height (long dimension) doubles the vertical collector dimension relative to landscape. The sensitive width increases from **2.29 m to 4.58 m**, and the module row casts a significantly longer shadow.

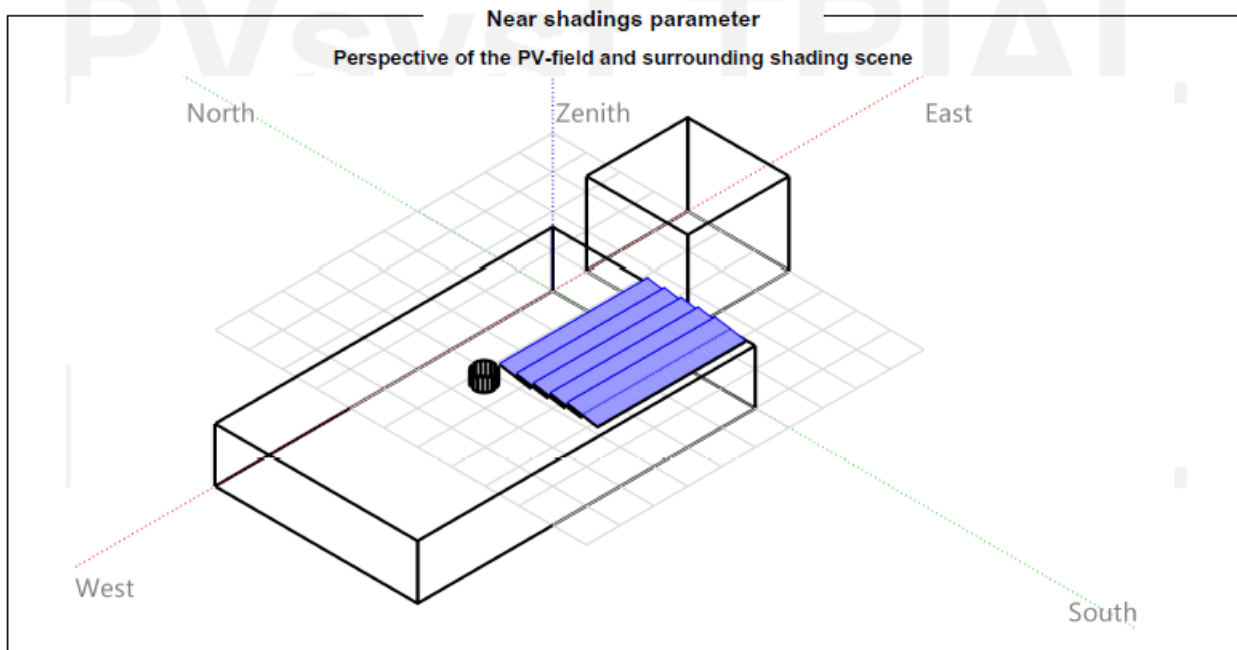


Figure 7: 3D perspective view showing portrait module arrangement

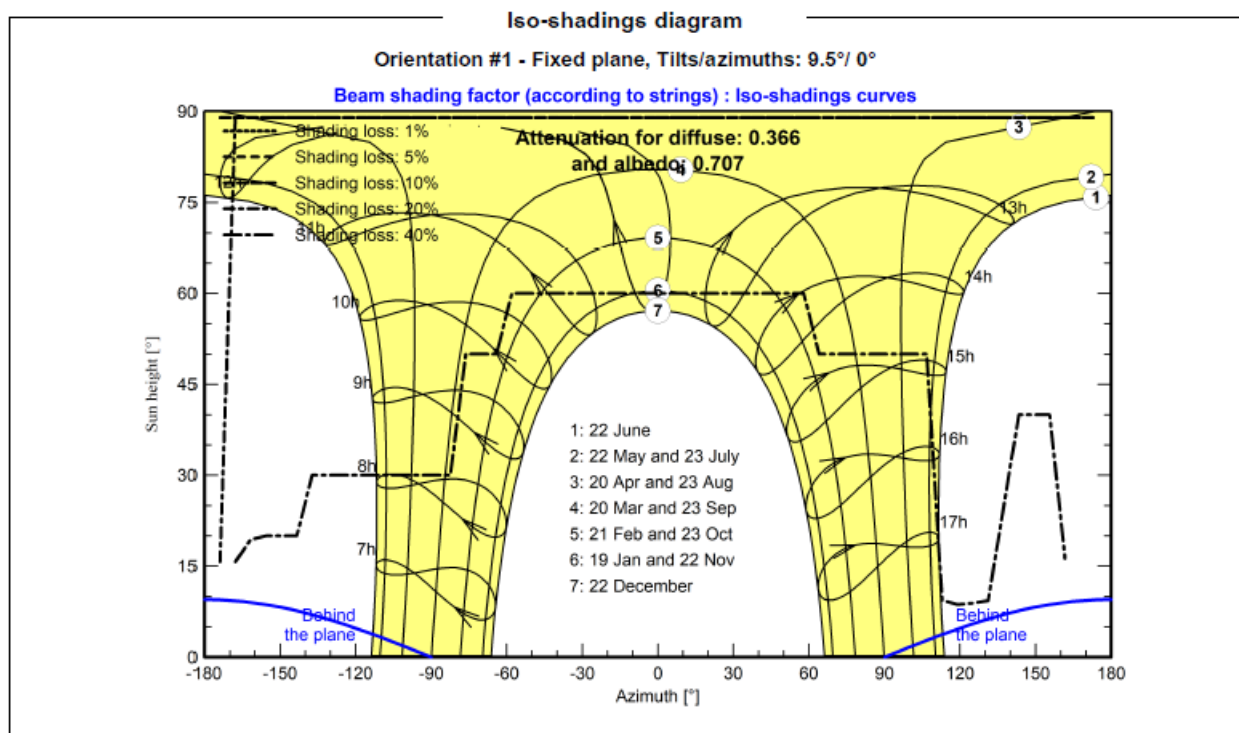


Figure 8: Iso-shading diagram

Table 5: Electrical Partition Results

Metric	Linear Baseline	Electrical Partition	Delta
Module Orientation	Landscape	Portrait	—
Sensitive Width	2.29 m	4.58 m	2× height
Near-Shading Irradiance Loss	−1.64%	−36.47%	−34.83 pp
IAM Factor Loss	−2.18%	−2.44%	−0.26 pp
Shadings: Electrical Loss (strings)	0.00%	−6.88%	−6.88 pp
Effective Global	1849.8 kWh/m ²	1191.5 kWh/m ²	−658.3 kWh/m ²
Array Virtual Energy at MPP	169,199 kWh	104,145 kWh	−65,054 kWh
Energy Injected to Grid	166,475 kWh	101,896 kWh	−64,579 kWh
Specific Production	1593 kWh/kWp	975 kWh/kWp	−618 kWh/kWp
Performance Ratio (PR)	82.86%	50.72%	−32.14 pp

Engineering Analysis: The portrait arrangement at 2.5 m pitch creates a geometric catastrophe. The module row height increases from ~1.13 m (landscape) to ~2.28 m (portrait), causing the front row to cast deep shadows onto the rear row during morning and afternoon hours. The near shading irradiance loss explodes from 1.64% to **36.47%**, reducing the effective global irradiance by 35.6%.

Beyond geometric loss, the **electrical partition model** applies an additional **6.88% penalty** due to string level current mismatch. In the Y=2, X=1 partition, each string spans the full 2 module height of the shed. When the bottom row shades the lower module of a string, the photocurrent in that substring collapses. Because modules in a series string must carry uniform current, the shaded substring limits the entire string. Bypass diodes across the half cell sections activate, short circuiting the shaded substring and reducing string voltage by approximately one-third. The inverter MPPT, tracking a P-V curve now fragmented into multiple local maxima, often locks onto a suboptimal peak. Since the two MPPT inputs have **no power sharing**, the unshaded MPPT cannot compensate for the current limited shaded MPPT, exacerbating the loss.

Quantification of Additional Electrical Mismatch: - Absolute electrical loss: 6.88% of 124,523 = **8,567 kWh**
 - Total yield penalty relative to baseline: 166,475 – 101,896 = **64,579 kWh (−38.8%)** - Of this penalty, ~36 percentage points are geometric (portrait height) and ~7 percentage points are electrical (mismatch + bypass diodes).

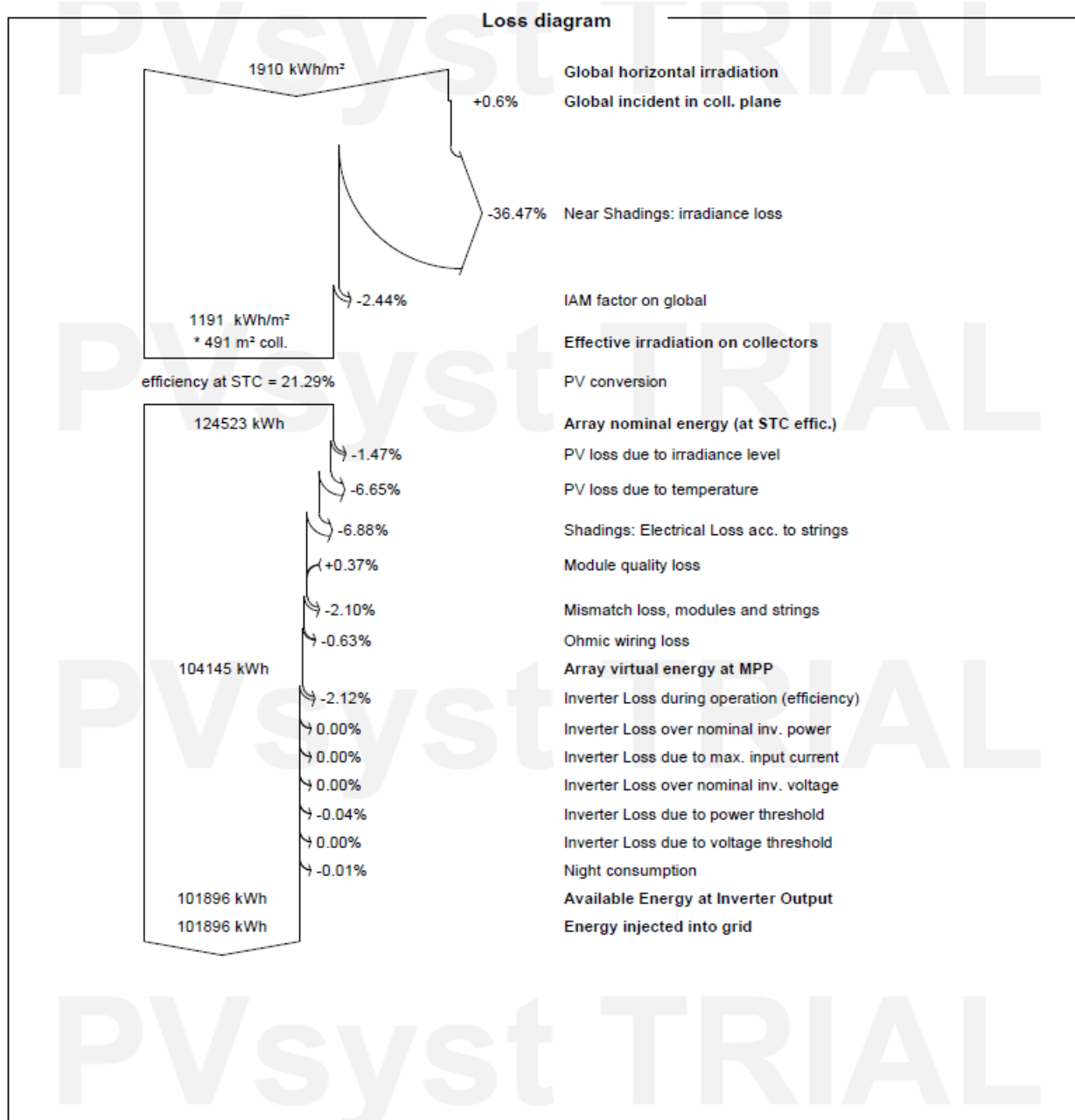


Figure 9: Loss Diagram

5. Exported Technical Diagrams

The following PVsyst exports are attached for technical documentation compliance: - **Loss Diagram** (Baseline and Electrical Partition variants) - **Iso-Shading Diagrams** (Linear and According to Strings) - **3D Layout Perspective View** - **Single-Line Diagram** (10 strings, 2 MPPTs, 100 kVA inverter)

6. Inter-Row Pitch Variation – 2.5 m → 3.2 m

Restoring **landscape** orientation and relaxing the inter-row pitch to **3.2 m** reduces the module packing density and lowers the shading limit angle.

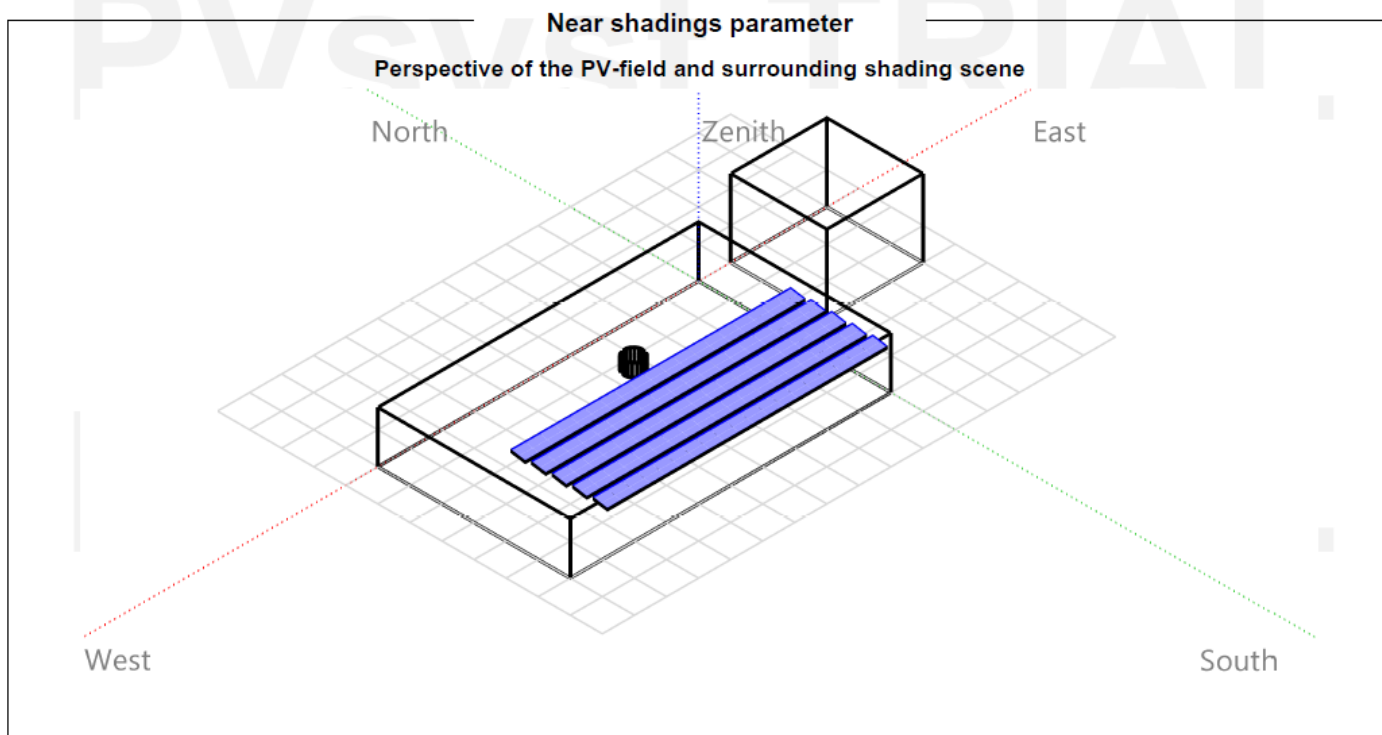


Figure 10: 3D perspective view with 3.2 m pitch

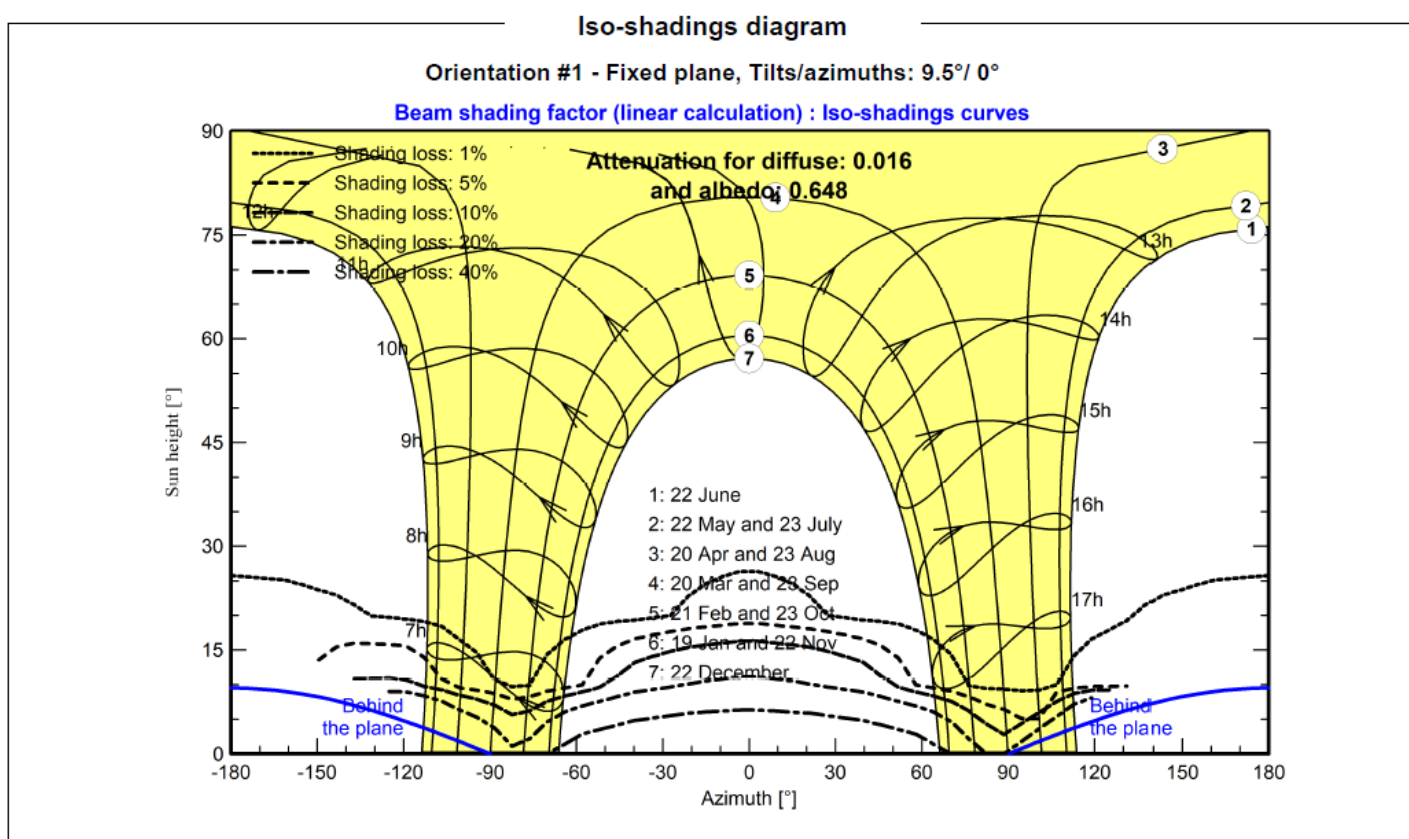


Figure 11: Iso shading diagram

Table 6: Pitch Variation Comparison (Landscape, 9.5° Tilt)

Metric	2.5 m Pitch	3.2 m Pitch	Improvement
GCR (Shading)	92.3%	72.1%	−20.2 pp
Shading Limit Angle	59.6°	22.4°	−37.2°
Near-Shading Loss	−1.64%	−0.61%	+1.03 pp
Effective Global	1849.8 kWh/m ²	1868.1 kWh/m²	+18.3 kWh/m ²
Array Virtual Energy	169,199 kWh	170,830 kWh	+1,631 kWh
Energy Injected to Grid	166,475 kWh	168,094 kWh	+1,619 kWh
PR	82.86%	83.67%	+0.81 pp
Specific Production	1593 kWh/kWp	1609 kWh/kWp	+16 kWh/kWp

Engineering Analysis: The 28% increase in pitch (0.7 m additional gap) yields a **1.0% energy gain** (+1,619 kWh/year) and improves PR by **0.81 percentage points**. The near shading loss is reduced by **63%** (from 1.64% to 0.61%). The GCR reduction to **72.1%** represents an optimal compromise for the 50 m × 30 m roof: the 5 sheds at 3.2 m pitch occupy only 16 m of roof length (plus setbacks), leaving adequate service corridors while maximizing yield per module. The limit profile angle of **22.4°** guarantees zero shading for all sun elevations above 22°, which covers >95% of daytime irradiance in Kilinochchi.

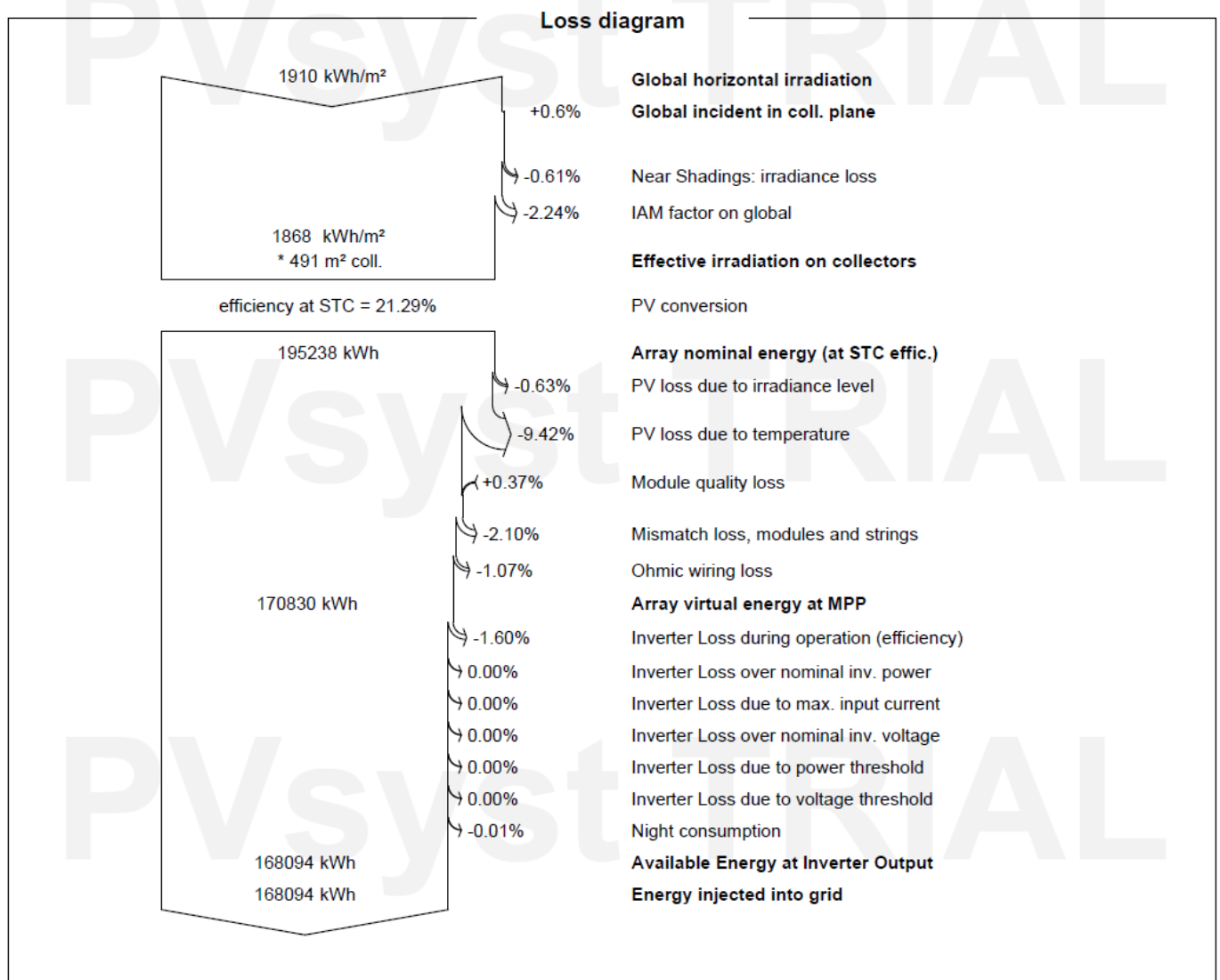


Figure 12: Loss Diagram

7. Tilt Angle Trade-Off – 9.5° → 18.0°

Maintaining the **3.2 m pitch** and **landscape** orientation, the array tilt is increased to **18.0°** to test the transposition-gain hypothesis.

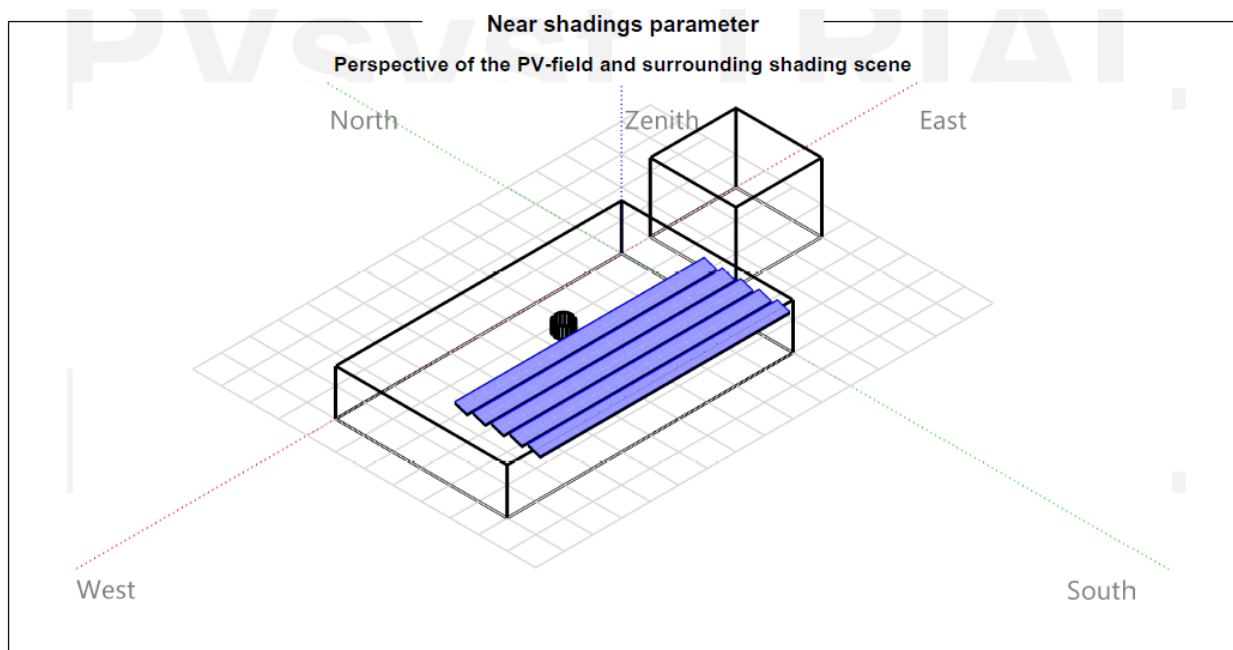


Figure 13: 3D perspective view with 18° tilt

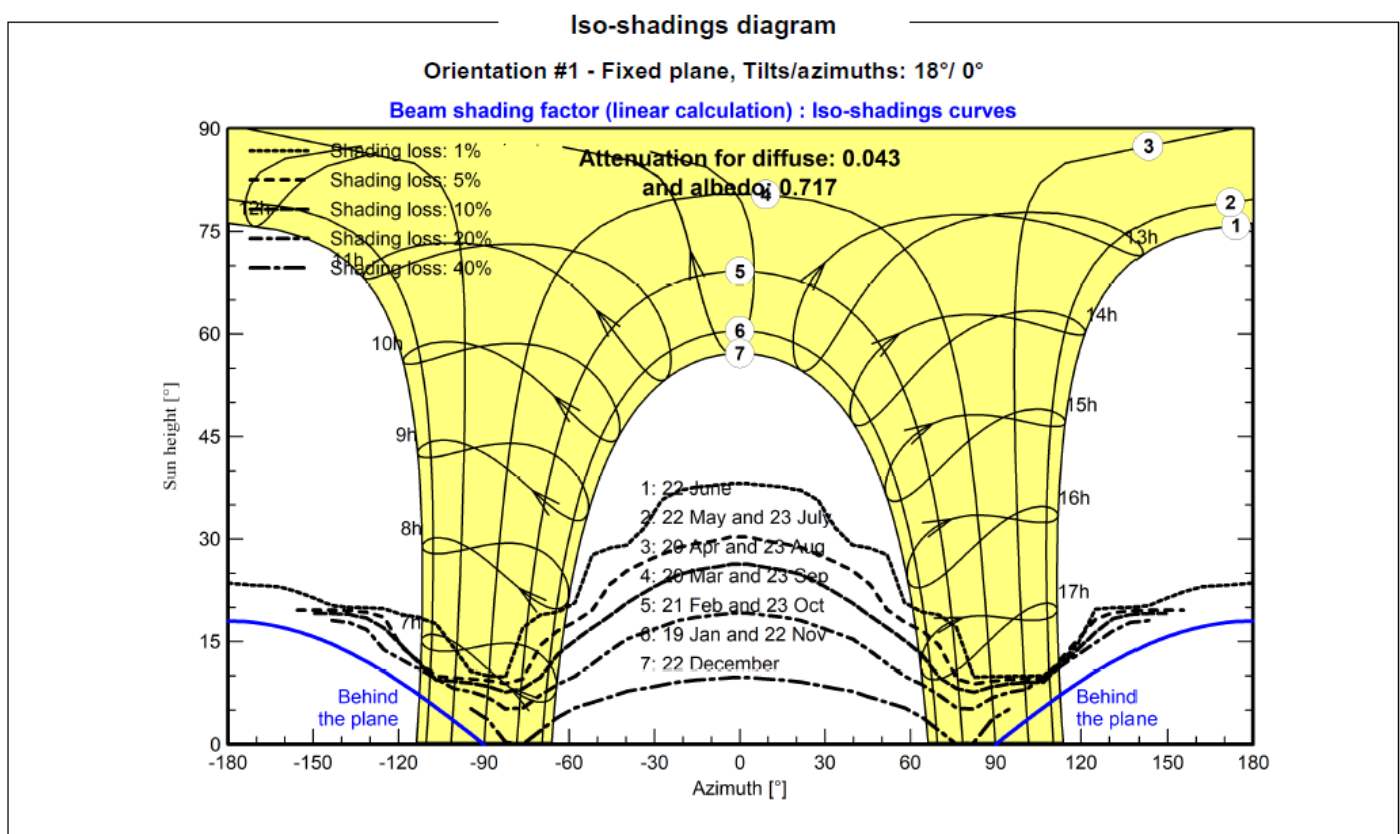


Figure 14: Iso shading diagram for 18°/0°

Table 7: Tilt Trade Off Comparison (Landscape, 3.2 m Pitch)

Metric	9.5° Tilt	18.0° Tilt	Delta
Tilt / Azimuth	9.5° / 0°	18.0° / 0°	+8.5°
Limit Profile Angle	22.4°	35.4°	+13.0°
Global Incident	1922.5 kWh/m ²	1899.8 kWh/m ²	-22.7 kWh/m ²
Near-Shading Loss	-0.61%	-1.68%	-1.07 pp
IAM Factor Loss	-2.24%	-2.18%	+0.06 pp
Effective Global	1868.1 kWh/m ²	1827.2 kWh/m ²	-40.9 kWh/m ²
Array Virtual Energy	170,830 kWh	167,261 kWh	-3,569 kWh
Energy Injected to Grid	168,094 kWh	164,552 kWh	-3,542 kWh
PR	83.67%	82.89%	-0.78 pp
Specific Production	1609 kWh/kWp	1575 kWh/kWp	-34 kWh/kWp

Engineering Analysis: At **9.4°N latitude**, the annual solar altitude is high; the theoretical optimal fixed tilt is often close to latitude (~9.5°). Increasing tilt to 18° introduces two competing effects:

1. **Transposition Gain:** A steeper tilt improves winter beam collection (lower solar altitude) and slightly increases diffuse capture from the sky dome.

2. **Inter-Row Shading Penalty:** The taller effective row height increases shadow length. The near-shading loss more than doubles ($0.61\% \rightarrow 1.68\%$), and the limit profile angle rises to **35.4°** , meaning shading persists for a larger fraction of the day.

For Kilinochchi, the **shading penalty dominates**. The global incident irradiance actually **decreases** from 1922.5 to 1899.8 kWh/m² because the steeper tilt rejects high-altitude midday beam irradiance (higher incidence angles near noon) and captures less ground-reflected albedo. Although IAM loss remains stable, the combined reduction in effective global irradiance (-40.9 kWh/m²) and increased self-shading causes E_{Grid} to drop by **3,542 kWh/year (-2.1%)**.

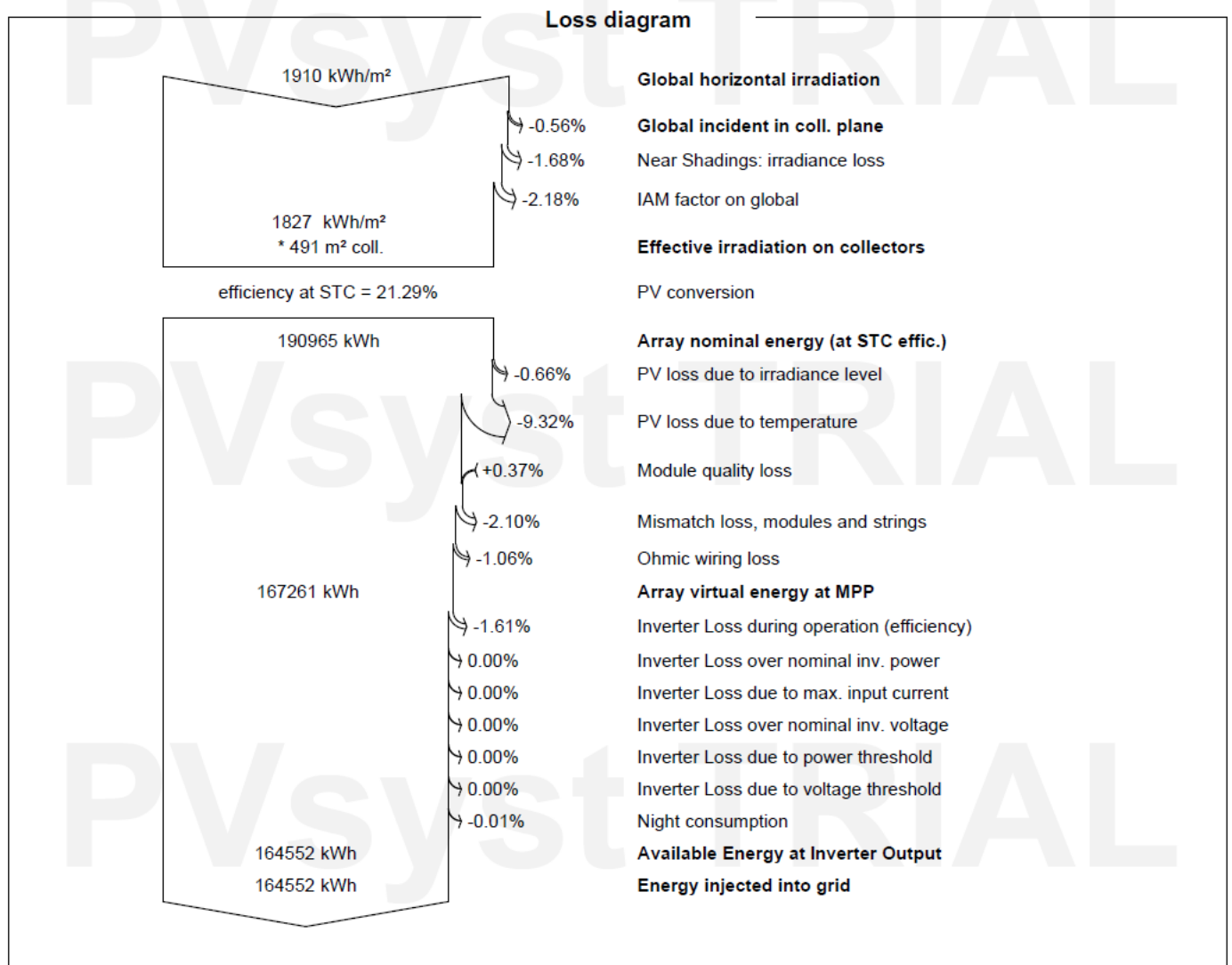


Figure 15: Loss Diagram

8. Synthesis of Variations Across Tasks 4, 6, and 7

- **Electrical Mismatch & Orientation (Task 4 vs. Task 2):** The 64,579 kWh (−38.8%) drop is dominated by the geometric shading increase from portrait module height (−36.47% linear loss) compounded by electrical string mismatch (−6.88%). The bypass diodes in half cell modules conduct when partial shading creates reverse bias, fragmenting the P-V curve and preventing the inverter from reaching the global MPP. The absence of MPPT power sharing locks the loss into the affected input.
- **Pitch Relaxation (Task 6 vs. Task 2):** The 1,619 kWh (+1.0%) gain results from reduced inter row shading (−1.64% → −0.61%). Wider pitch allows more diffuse and low angle beam irradiance to reach the lower portion of each row during mornings and afternoons. The energy per-module improves even though the roof power density (W/m²) decreases.
- **Tilt Aggressiveness (Task 7 vs. Task 6):** The 3,542 kWh (−2.1%) drop at 18° tilt demonstrates the **tropical tilt paradox**. While transposition gain theoretically favours tilt angles closer to latitude, the high annual solar altitude at 9.4°N means steeper tilt increases reflection and self shading without sufficient compensating beam gain. The 18° tilt is only justified for winter peaking load profiles; for uniform annual grid injection, **9.5° is superior**.

DISCUSSION

1. Ground Coverage Ratio (GCR) and Inter Row Pitch in Tropical Solar Geometries

GCR quantifies the fraction of roof area covered by active collector surface:

$$GCR = \frac{\text{Total Module Area}}{\text{Roof Footprint Area}} = \frac{491 \text{ m}^2}{1500 \text{ m}^2} \approx 32.7\%$$

However, PVsyst reports a **shading GCR** metric that relates the sensitive collector width to the row pitch. For landscape modules (~1.13 m width) at 2.5 m pitch, the shading GCR of 92.3% indicates that the collector width occupies 92.3% of the pitch distance a dense packing. At 3.2 m pitch, this drops to 72.1%, reflecting the relaxed spacing.

In tropical geometries (latitude < 15°), the sun path is nearly vertical at solar noon (zenith angle $\approx 0^\circ$ during equinoxes). This creates a unique design envelope:

- **Low tilt advantage:** A near flat array captures maximum beam irradiance at midday with $\cos(\theta) \approx 1$ and minimal reflection.
- **Shading sensitivity:** Because the sun is high, row to row shading occurs primarily during mornings and afternoons when the azimuth is oblique. However, with tall modules (portrait) or steep tilt, the shadow length increases dramatically according to:

$$\tan(\theta_{limit}) = \frac{\text{Pitch} \times \cos(\alpha_{az})}{\text{Row Height}}$$

For Kilinochchi, the 3.2 m pitch with landscape modules provides a limit angle of **22.4°**, ensuring no shading for sun elevations above 22° throughout the year. The 2.5 m pitch raises this to 59.6°, accepting minor shading in exchange for higher module count per roof. For rooftop projects where roof area is fixed, GCR optimization becomes a trade off between maximizing installed capacity (high GCR) and maximizing specific yield (low GCR). The simulation results indicate that for this 100 kWp system, the **3.2 m pitch (GCR 72.1%) achieves the highest PR (83.67%)** and should be preferred unless roof area constraints force a denser layout.

1. Electrical Partition Shading Losses and Bypass Diode I-V Curve Mechanics

Electrical partition losses are consistently higher than linear shading because linear methods assume uniform current reduction proportional to irradiance. In reality, modern PV modules contain **bypass diodes** (typically one per half-cell substring) to prevent hot spots under partial shading.

When a series string experiences partial shading: - The shaded cells operate in reverse bias, limiting string current to the shaded substring's photocurrent. The bypass diode across the shaded substring activates once the reverse bias exceeds the diode threshold ($\sim 0.6\text{--}0.7\text{ V}$), short circuiting that section and reducing the string voltage by approximately half (for half-cell modules with 3 diodes). The inverter MPPT, tracking the global maximum, now faces a P-V curve with **multiple local peaks**. If the global peak lies on a lower voltage bypass branch, the array operates at reduced voltage and current. The power loss is **non linear** with shaded area: a small shaded fraction can cause a disproportionately large current drop.

In PVsyst's electrical partition model, the portrait Y=2, X=1 partition means each string spans the full 2 module height of the shed. When the bottom row shades the lower module of the next shed's string, that entire string's current collapses to the shaded substring level. The **-6.88% electrical loss** represents the additional power penalty beyond the geometric irradiance reduction (-36.47%) due to this mismatch and bypass diode conduction.

The I-V curve mechanics explain why module orientation matters: **landscape modules** (shorter height) reduce the probability that a row-to-row shadow falls across the height of a single electrical substring, whereas **portrait modules** (taller height) increase this probability. This is why commercial utility-scale projects in flat-roof tropical markets overwhelmingly adopt landscape orientation with low tilt.

2. PVsyst Loss Tree Parameters and Field Maintenance Strategies

The PVsyst Loss Tree decomposes energy flow from horizontal irradiation to grid injection. Key parameters and their long-term degradation mechanisms:

Thermal Loss Factor ($U_c = 20.0 \text{ W/m}^2\text{K}$): The thermal model translates irradiance into cell temperature via the one diode heat balance equation. At 50°C operating conditions, the Generic 550 Wp module loses ~9.4% efficiency relative to STC (25°C). In Kilinochchi's tropical climate (annual ambient ~28°C), modules routinely operate at 60–70°C. A soiled module has reduced emissivity and convective heat transfer, increasing U_c and exacerbating temperature losses.

Maintenance strategy: Ensure adequate roof ventilation beneath modules; implement a 6 week cleaning cycle to maintain heat dissipation.

Ohmic Wiring Loss (1.07% at STC, 88 mΩ): DC resistance in 4 mm² Cu cables causes I^2R loss. Over time, connector corrosion, loose terminations, and insulation degradation increase resistance. A 20% increase in contact resistance at high irradiance hours can raise ohmic loss to >1.5%.

Maintenance strategy: Annual thermographic inspection of string combiners and DC cabling to detect hot joints; re torque MC4 connectors every 2 years.

Mismatch Loss (2.10%): Factory power tolerance (typically ±3%), uneven soiling, and partial shading cause current-voltage mismatch between modules and strings. Over time, PID (Potential Induced Degradation), LID (Light Induced Degradation), and cell micro-cracking increase mismatch.

Maintenance strategy: Electroluminescence (EL) testing at commissioning to identify cracked cells; replace modules showing >5% power degradation after 5 years.

Soiling Factor (embedded in environmental/model settings): In Kilinochchi's dry tropical climate, dust accumulation can reduce irradiance by 2–5% monthly during dry seasons (March–September). Unlike temperature degradation, soiling is fully recoverable.

Maintenance strategy: Implement automated or manual cleaning on a 6-week cycle during dry months. A 2% soiling loss on 168 MWh/year equals **3.36 MWh lost energy** worth ~\$500–800 annually, easily justifying cleaning labour costs.

Long-Term PR Degradation Trajectory: A new system PR of 83.67% (Task 6) will degrade by approximately 0.5–0.8 percentage points per year without maintenance: - **Year 1–2:** LID causes ~1.5% power drop; PR falls to ~82%. - **Year 3–5:** Soiling accumulation and connector oxidation add 1–2%; PR falls to ~80%. - **Year 5–10:** PID, encapsulant browning, and bypass diode failures accelerate; PR may drop below 78%.

Preventive maintenance cleaning, torque checks, and IV curve tracing can limit degradation to **0.3% per year**, preserving 80%+ PR for 10 years. The PVsyst Loss Tree provides the quantitative baseline against which field measurements are benchmarked.